

Forward operation of adenine nucleotide translocase during F₀F₁-ATPase reversal: critical role of matrix substrate-level phosphorylation

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ABSTRACT In pathological conditions, F₀F₁-ATPase hydrolyzes ATP in an attempt to maintain mitochondrial membrane potential. Using thermodynamic assumptions and computer modeling, we established that mitochondrial membrane potential can be more negative than the reversal potential of the adenine nucleotide translocase (ANT) but more positive than that of the F₀F₁-ATPase. Experiments on isolated mitochondria demonstrated that, when the electron transport chain is compromised, the F₀F₁-ATPase reverses, and the membrane potential is maintained as long as matrix substrate-level phosphorylation is functional, without a concomitant reversal of the ANT. Consistently, no cytosolic ATP consumption was observed using plasmalemmal K_{ATP} channels as cytosolic ATP biosensors in cultured neurons, in which their *in situ* mitochondria were compromised by respiratory chain inhibitors. This finding was further corroborated by quantitative measurements of mitochondrial membrane potential, oxygen consumption, and extracellular acidification rates, indicating nonreversal of ANT of compromised *in situ* neuronal and astrocytic mitochondria; and by bioluminescence ATP measurements in COS-7 cells transfected with cytosolic or nuclear-targeted luciferases and treated with mitochondrial respiratory chain inhibitors in the presence of glycolytic plus mitochondrial *vs.* only mitochondrial substrates. Our findings imply the possibility of a rescue mechanism that is protecting against cytosolic/nuclear ATP depletion under pathological conditions involving impaired respiration. This mechanism comes into play when mitochondria respire on substrates that support matrix substrate-level phosphorylation.—Chinopoulos, C., Gerencser, A. A., Mandi, M., Mathe, K., Töröcsik, B., Doczi, J., Turiak, L., Kiss, G., Konrád, C., Vajda, S., Vereczki, V., Oh, R. J., Adam-Vizi, V. Forward operation of adenine nucleotide translocase during F₀F₁-ATPase reversal: critical role of matrix substrate-level phosphorylation. *FASEB J.* 24, 2405–2416 (2010). www.fasebj.org

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MITOCHONDRIAL ATP SYNTHASE (F₀F₁-ATPase) is able to synthesize and to hydrolyze ATP (1–5). Because nucleotide phosphorylation/dephosphorylation takes place on the matrix side of the ATP synthase, obviously only ADP and ATP present in the matrix participate in the production/consumption process. Extramitochondrial adenine nucleotides can contribute to the matrix adenine nucleotide pool only through the adenine nucleotide translocase (ANT) and, by a minor fraction, through the ATP-Mg/P_i carrier (6). Both ANT and F₀F₁-ATPase catalyze reversible processes; their directionality is governed by the mitochondrial membrane potential (ΔΨ_m) and their respective “reversal potential” (*E*_{rev}), the latter determined by the concentrations of the participating reactants. For ANT, this is referred to as *E*_{rev_ANT} and for the F₀F₁-ATPase as *E*_{rev_ATPase} and the values (mV) are given by Eqs. 1 and 2:

$$E_{\text{rev_ANT}} = \frac{2.3RT}{F} \cdot \log \frac{[\text{ADP}^{3-}]_{\text{free_out}} \cdot [\text{ATP}^{4-}]_{\text{free_in}}}{[\text{ADP}^{3-}]_{\text{free_in}} \cdot [\text{ATP}^{4-}]_{\text{free_out}}} \quad (1)$$

$$E_{\text{rev_ATPase}} = -\frac{316}{n} - \frac{2.3RT/F}{n} \times \log \frac{[\text{ATP}^{4-}]_{\text{free_in}}}{[\text{ADP}^{3-}]_{\text{free_in}} \cdot [\text{P}^-]_{\text{in}}} - \frac{2.3RT}{F} \cdot (\text{pH}_o - \text{pH}_i) \quad (2)$$

where o or out is outside the matrix; i or in is inside the matrix; *n* is the H⁺/ATP coupling ratio; *R* is the universal gas constant, 8.31 J · mol^{−1} · K^{−1}; *F* is the Faraday constant, 9.64 × 10⁴ C · mol^{−1}; *T* is temperature (K); and [P[−]] is the free phosphate concentration given by Eq. 3:

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$$[P^-]_{in} = \frac{[P_{total}]_{in}}{1 + 10^{pHi - pKa2}} \quad (3)$$

where $pK_{a2} = 7.2$ for phosphoric acid.

When $\Delta\Psi_m$ is more negative than E_{rev_ATPase} and E_{rev_ANT} , ATP is synthesized in the matrix and ANT operates in the “forward” mode, translocating ADP into and ATP out of the matrix. With $\Delta\Psi_m$ being less negative than E_{rev_ATPase} and E_{rev_ANT} , ATP is hydrolyzed and ANT operates in the “reverse” mode, translocating ATP into and ADP out of the matrix.

It is apparent from Eqs. 1 and 2 that alterations in $[ATP^{4-}]_{in}$ and $[ATP^{3-}]_{in}$ alter E_{rev_ATPase} and E_{rev_ANT} . In fact, these would change inversely; a decrease in the $[ATP^{4-}]_{in}/[ATP^{3-}]_{in}$ ratio would shift E_{rev_ATPase} toward more positive and E_{rev_ANT} toward more negative values and *vice versa*. The $[ATP^{4-}]_{in}/[ATP^{3-}]_{in}$ ratio is a variable element, being dependent on the current metabolic demand under physiological conditions. In addition, other parameters may also exhibit wide fluctuations: phosphate (7), the coupling ratio n (8, 9), and $\Delta\Psi_m$ ($\Delta\Psi_m$ flickering), the latter of which could be greater than 100 mV (10–12). The operation of ANT in intact isolated mitochondria has been modeled recently (13).

A decrease in $\Delta\Psi_m$ due to electron transport chain (ETC) inhibition or to an increase in the inner membrane permeability stops ATP synthesis and allows the ATP synthase to reverse (reviewed in ref. 14), leading to bioenergetic failure in cells. Depletion of ATP produced in glycolysis due to, *e.g.*, an overly stimulated plasmalemmal Na^+/K^+ ATPase is a crucial element in the development of energy crisis, which should involve the reverse operation of ANT. However, it is a realistic assumption that $\Delta\Psi_m$ could be in a range where ANT and F_0F_1 -ATPase function in opposite directions, in particular, under pathological conditions involving a decrease in the $[ATP^{4-}]_{in}/[ADP^{3-}]_{in}$ ratio. Therefore, the question arises whether mitochondria are able to hydrolyze matrical ATP without the assistance of ANT. In this respect, alternative ATP sources potentially available in mitochondria are to be considered.

In the mitochondrial matrix, two reactions are capable of substrate-level phosphorylation: the mitochondrial phosphoenolpyruvate carboxykinase (PEPCK) and the succinate-CoA ligase (SUCL; or succinate thiokinase or succinyl-CoA synthetase). Mitochondrial PEPCK is thought to participate in the transfer of the phosphorylation potential from the matrix to cytosol and *vice versa* (15–18). SUCL catalyzes the reversible conversion of succinyl-CoA and ADP or GDP to CoASH, succinate, and ATP or GTP (19). As such, it plays a key role in the citric acid cycle (20), ketone metabolism (16), and heme synthesis (21), as well as being a phosphate target for the activation of mitochondrial metabolism (22). The enzyme is a heterodimer, being composed of an invariant α subunit encoded by *SUCLG1* and a substrate-specific β subunit, encoded by either *SUCLA2* or *SUCLG2*. This dimer combination results in either an ADP-forming SUCL (EC 6.2.1.5) or a GDP-forming SUCL (EC 6.2.1.4).

In the present work, we addressed the function of F_0F_1 -ATPase while shifting E_{rev_ATPase} and E_{rev_ANT} to different directions, using a broad range of experiments and modeling; specifically, 1) a computer model was developed to estimate E_{rev_ATPase} and E_{rev_ANT} at different bioenergetic parameters, 2) experiments were performed on isolated and on *in situ* mitochondria demonstrating a dissociation between E_{rev_ATPase} and E_{rev_ANT} and its consequences on cytosolic and nuclear ATP levels, and 3) the contribution of substrate-level phosphorylation to the directionality of F_0F_1 -ATPase and ANT was determined.

The results highlight the fact that mitochondria could hydrolyze matrix-produced ATP in a $\Delta\Psi_m$ range more positive than E_{rev_ATPase} but more negative than E_{rev_ANT} , that is, when ANT still functions in the forward mode. This mechanism, which is efficient for as long as substrate-level phosphorylation provides ATP for F_0F_1 -ATPase, has an important implication for pathology; because in this $\Delta\Psi_m$ range, the extramitochondrial ATP reserves are spared, providing better chances for cell survival.

MATERIALS AND METHODS

Isolation of mitochondria from rat and rabbit livers and hearts

Mitochondria from rat and rabbit livers and hearts were isolated in a manner identical to that detailed previously (23), with minor modifications as discussed in ref. 24. Male Sprague-Dawley rats weighing 300–350 g and male New Zealand albino rabbits weighing 2–2.5 g were used. All animal procedures were performed according to the local animal care and use committee (Egyetemi Allatkiszerleti Bizottság) guidelines.

Determination of $[Mg^{2+}]_{free}$ by Magnesium Green fluorescence in the extramitochondrial volume of isolated mitochondria and conversion of $[Mg^{2+}]_{free}$ to the ADP-ATP exchange rate mediated by the ANT

The ADP-ATP exchange rate was estimated using a method described recently by our laboratory (24), exploiting the differential affinity of ADP and ATP to Mg^{2+} . The rate of ATP appearing in the medium after addition of ADP to energized mitochondria (or *vice versa* in the case of deenergized mitochondria) is calculated from the measured rate of change in free extramitochondrial $[Mg^{2+}]$ using standard binding equations. The assay is designed such that the ANT is the sole mediator of changes in $[Mg^{2+}]_{free}$ in the extramitochondrial volume, as a result of ADP-ATP exchange. For the calculation of [ATP] or [ADP] from $[Mg^{2+}]_{free}$, the apparent K_d values are identical to those in ref. 24 owing to identical experimental conditions ($K_{ADP}=0.906\pm0.023$ mM; $K_{ATP}=0.114\pm0.005$ mM).

Determination of $\Delta\Psi_m$ in isolated mitochondria

$\Delta\Psi_m$ was estimated using fluorescence quenching of the cationic dye safranin O, owing to its accumulation inside energized mitochondria (25), and calibrated as described elsewhere (24).

Preparation of cortical neurons

Primary cultures of cortical neurons were prepared from Sprague-Dawley rats (d 17 *in utero*) as described elsewhere (26). All animal procedures were performed according to the local guidelines.

COS-7 cell cultures and transfection with the luciferase plasmids

Luciferases targeting the cytosol or nucleus were constructed by Manfredi's group (27). COS-7 cells were grown on 175-cm² flasks in DMEM with glutamine, 10% FCS, and 1% streptomycin-penicillin. On reaching confluence (15–17 × 10⁶ cells/flask), cultures were harvested by trypsinization and were transfected by electroporation according to the manufacturer's instructions (Amaza Inc., Gaithersburg, MD, USA). In brief, 3 µg of cytosol- or nuclear-targeted luciferase plasmid was electroporated into 5 × 10⁶ cells in 100 µl of electroporation medium. Then, ~70,000 transfected cells were seeded per 25-mm round coverslip precoated with poly-L-ornithine. Transfection efficiency was in the 25–40% range, as verified by immunocytochemistry using a monoclonal antibody raised against luciferase protein from *Photinus pyralis* (10 µg of primary mouse antibody, clone CS 17, IgG1 isotype; Invitrogen, Carlsbad, CA, USA). Images were evaluated using a 2100 Rainbow Confocal System (Bio-Rad Laboratories, Hercules, CA, USA).

Measurement of oxygen consumption and extracellular acidification rates in cultured cortical neurons

Oxygen consumption rate (OCR) and extracellular acidification rate (ECAR) measurements were performed on d 11–13 of culture in an XF24 Extracellular Flux Analyzer (Seahorse Bioscience, North Billerica, MA, USA) in a low-buffering-capacity medium containing 120 mM NaCl, 3.5 mM KCl, 1.3 mM CaCl₂, 1 mM MgCl₂, 0.4 mM KH₂PO₄, 1.2 mM Na₂SO₄, and 15 mM glucose at pH 7.4 at 37°C, as detailed previously (28). To prevent excitotoxicity all experiments were performed in the presence of tetrodotoxin (1 µM), MK801 (10 µM), 2,3-dihydroxy-6-nitro-7-sulfamoylbenzo(*f*)quinoxaline (10 µM), and nifedipine (1 µM).

In situ determination of ΔΨ_m in neuronal and astrocytic mitochondria

Cortical cultures were grown in Lab-Tek 8-well chambered coverglasses (Nunc, Rochester, NY, USA). ΔΨ_m was measured by wide-field fluorescence imaging, as described in Supplemental Material. In brief, the fluorescence of the mitochondrial potentiometric fluorescent dye tetramethylrhodamine methyl ester (TMRM) (7.5 nM) and an anionic plasma membrane potential indicator (PMPI) were followed in time over cell bodies. ΔΨ_m was calibrated to millivolts by modeling the redistribution of TMRM across the mitochondrial and plasma membranes (29) and PMPI across the plasma membrane (30). Experiments were performed at 37°C in the medium used for OCR measurements above, supplemented by NaHCO₃ (5 mM), *N*-tris(hydroxymethyl)methyl-2-aminoethanesulfonic acid (20 mM), and tetraphenylboron (1 µM). Calibration of ΔΨ_m from raw TMRM fluorescence is described in Supplemental Material.

Fluorescence imaging

Imaging was performed on an Olympus IX-81 inverted microscope (Olympus, Tokyo, Japan) equipped with a UAPO ×20 air 0.75NA lens, a Lambda LS Xe-arc light source (175 W), Lambda 10-2 excitation and emission filter wheels (Sutter Instruments, Novato, CA, USA), a ProScan linear encoded

xy-stage (Prior, Rockland, MA, USA), and a CoolSNAP HQ cooled digital charge-coupled device camera (−30°C, 10 MHz readout, low gain, 12-bit depth; Photometrics, Tucson, AZ, USA). Full frames at 4 × 4 binning (1.3 µm/pixel resolution) were taken at 3-min intervals. Twelve to 30 view fields in 4 to 6 wells of the Lab-Tek chamber were recorded cyclically using the Multi Dimensional Acquisition feature of Metamorph 6.3 (Molecular Devices, Downingtown, PA, USA). The filter sets, given as excitation-dichroic mirror-emission (nm) were for PMPI 500/24–520LP (long pass)-542/27 (all from Semrock, Rochester, NY, USA) and for TMRM 555-475/560LP-620/40 (all from Chroma, Rockingham, VT, USA). The cross-talk of TMRM and PMPI emissions was eliminated by linear spectral unmixing of intensity traces in Mathematica 5.2 as described previously (30).

Luciferase assay in COS-7 cells

For luciferase assays, 25-mm round coverslips with attached cells (transfected as detailed above) were incubated for 3 min in a medium containing 120 mM NaCl, 3.5 mM KCl, 20 mM HEPES-Na, 1 mM MgCl₂, 1.3 mM CaCl₂, 15 mM glucose (where indicated), 5 mM pyruvate sodium (where indicated, together with 2 mM 2-deoxyglucose), and 250 µM iodoacetamide, pH 7.38. Mitochondrial poisons and luciferin (1 mM) were added, and luminescence was measured in a Victor³ Wallace plate reader luminometer (PerkinElmer Life and Analytical Sciences, Boston, MA, USA). Luminescence signals were recorded over a 60-min time interval, and results are shown as the area under the curve of the luminescence intensity.

Western blotting

Immunoblotting of solubilized mitochondria was performed as recommended by the manufacturers of the antibodies. Rabbit polyclonal antibody to succinate-CoA ligase, ADP-forming, β subunit (Proteintech Europe Ltd, Manchester, UK), the primary antibody, was used at 1:1000. Mouse monoclonal antibody to VDAC-1 (Calbiochem, San Diego, CA, USA) was used at 1:20,000. Immunoreactivity was detected using peroxidase-linked secondary antibodies (1:10,000; Jackson ImmunoResearch Laboratories, West Grove, PA, USA) and enhanced chemiluminescence detection reagent (ECL system; Amersham Biosciences GE Healthcare Europe GmbH, Vienna, Austria).

Cell-attached electrophysiological recordings

Patch pipettes were pulled from borosilicate glass to resistances of 6–8 MΩ. The pipette solution contained 120 mM KCl, 20 mM HEPES acid, 4 mM MgCl₂, 1.3 mM CaCl₂, 5 mM EGTA (pH 7.38), 1 µM tetrodotoxin, 12 µM MK801, 10 µM 6-cyano-2,3-dihydroxy-7-nitroquinoxaline (CNQX), 1 µM nimodipine, 10 nM charybdotoxin, 10 mM iberiotoxin, 0.5 µM apamin, 1 µM triaryl methane-34, and 100 nM clotrimazole (pH 7.4). Bath solutions contained 120 mM NaCl, 3.5 mM KCl, 20 mM HEPES-Na, 1 mM MgCl₂, 1.3 mM CaCl₂, 15 mM glucose (pH 7.38), 1 µM tetrodotoxin, 12 µM MK801, 10 µM CNQX, 1 µM nimodipine, 250 µM iodoacetamide, and 100 µM pinacidil. Cultured neurons were placed into a chamber and visualized under differential interference contrast in a nonperfused manner. Seal resistances of >10 GΩ were routinely obtained, and patches remained stable for tens of minutes. Currents were acquired at 1 kHz, amplified using an Axopatch 200B amplifier (Axon Instruments, Foster City, CA, USA), digitized at a sampling rate of 10 kHz (Digidata 1322A; Axon

Instruments), and recorded using pCLAMP 10 software (Axon Instruments). All recordings were done at ambient room temperature ($\sim 23^{\circ}\text{C}$).

Reagents

Standard laboratory chemicals and reagents unless otherwise stated were from Sigma-Aldrich (St. Louis, MO, USA). Magnesium Green and TMRM were from Invitrogen. PMPI (R8042 FLIPR Membrane Potential Assay Kit) was from Molecular Devices (Sunnyvale, CA, USA). SF 6847 [also known as Tyrphostin 9, RG-50872, Malonaben, 3,5-di-*tert*-butyl-4-hydroxybenzylidenemalononitrile, and 2,6-di-*t*-butyl-4-(2',2'-dicyanovinyl)phenol; catalog number EI-215] was from Biomol GmbH (Hamburg, Germany). Carboxyatractyloside (cATR) was from Calbiochem. DMEM, RPMI 1640, Neurobasal medium, B27 supplement, and Alexa 647 lectin conjugate were from Invitrogen. All mitochondrial substrate stock solutions were dissolved in bidistilled water and titrated to pH 7.0. ADP and ATP were purchased as K^+ salts of the highest purity available and titrated to pH 6.9.

Statistics

Data are presented as means $\pm$ SE; significant differences between groups of data were evaluated by 1-way ANOVA followed by Tukey's *post hoc* analysis, with $P < 0.05$ considered as significant. Wherever single graphs are presented, they are representative of ≥ 4 independent experiments.

RESULTS

Computational estimations of $E_{\text{rev_ATPase}}$ and $E_{\text{rev_ANT}}$

From Eqs. 1 and 2, the concentration of free ATP^{4-} ($[L]$) can be obtained by Eq. 4:

$$[L] = \frac{[L]_t / (1 + [\text{Mg}^{2+}]_{\text{free}} / K_{\text{M,app}})}{1 + 10^{(-\text{pH})} / K_{\text{H}}} \quad (4)$$

where $[L]_t$ is the total ATP concentration, *i.e.*, $[\text{ATP}^{4-}] + [\text{ATP-H}^{3-}] + [\text{ATP-Mg}^{2-}] + [\text{ATP-H-Mg}^-]$; K_{H} is the dissociation constant for the reaction $\text{ATP-H}^{3-} \leftrightarrow \text{ATP}^{4-} + \text{H}^+$; and $K_{\text{M,app}}$ is the apparent dissociation constant of MgATP measured at pH 7.25 and 37°C (24). Similarly, the concentration of free ADP^{3-} can also be obtained using Eq. 4 ($[L]$, free $[\text{ADP}^{3-}]$; $[L]_t$, total ADP concentration, *i.e.*, $[\text{ADP}^{3-}] + [\text{ADP-H}^{2-}] + [\text{ADP-Mg}^-] + [\text{ADP-H-Mg}]$; $K_{\text{M,app}}$, apparent dissociation constant of MgADP measured at pH 7.25 and 37°C ; ref. 24). The values for K_{H} and $K_{\text{M,app}}$ might be difficult to determine for the conditions found inside the matrix, and we assume that they are not different from those found outside the matrix. Because the matrix $[\text{Mg}^{2+}]$ and pH_i are only slightly different from the extramitochondrial $[\text{Mg}^{2+}]$ and pH_o (24), it is assumed that both ANT and the $\text{F}_0\text{F}_1\text{-ATPase}$ handle the measured free $[\text{ATP}]$ and $[\text{ADP}]$. In Fig. 1 we computed $E_{\text{rev_ANT}}$ (white triangles) and $E_{\text{rev_ATPase}}$ (black triangles) for a range of free $[\text{ATP}]_{\text{in}}$ and $[\text{ADP}]_{\text{in}}$ as an input to Eqs. 1 and 2. It follows from Fig. 1 that during progressive depolarization of mitochondria, $\Delta\Psi\text{m}$ could be in a range (B in Fig. 1) where the

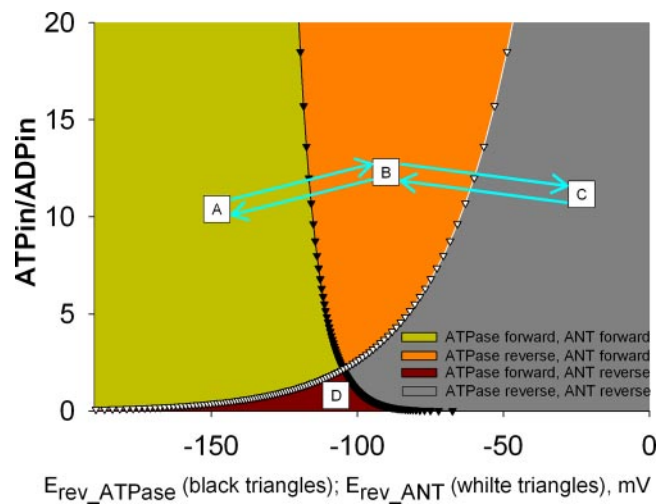
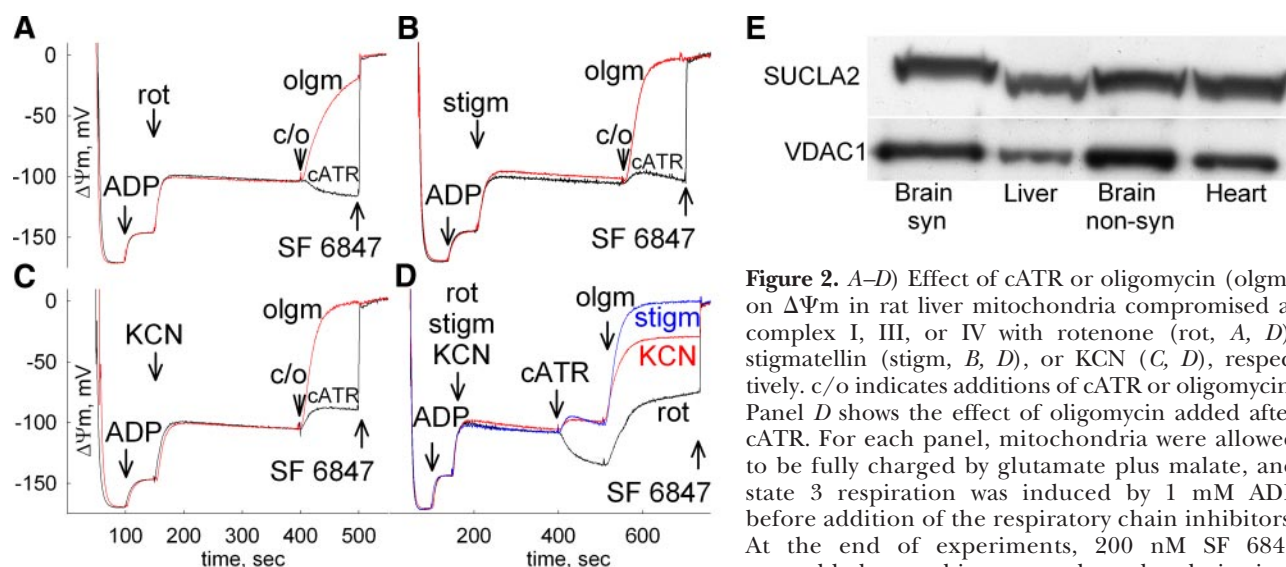


Figure 1. Computational estimation of $E_{\text{rev_ANT}}$ and $E_{\text{rev_ATPase}}$ at different free $[\text{ATP}]_{\text{in}}/[\text{ADP}]_{\text{in}}$ ratios. A: ATPase forward, ANT forward; B: ATPase reverse, ANT forward; C: ATPase reverse, ANT reverse; D: ATPase forward, ANT reverse. Black triangles represent $E_{\text{rev_ATPase}}$; white triangles represent $E_{\text{rev_ANT}}$. Values were computed for $[\text{ATP}]_{\text{out}} = 1.2 \text{ mM}$, $[\text{ADP}]_{\text{out}} = 10 \text{ }\mu\text{M}$, $P_{\text{in}} = 0.02 \text{ M}$, $n = 3.9$, $\text{pH}_i = 7.38$, and $\text{pH}_o = 7.25$.

$\text{F}_0\text{F}_1\text{-ATPase}$ is reversed, but ANT still functions in the forward mode. For the estimations, other parameters were assumed to be static, specifically those of $[\text{ATP}]_{\text{out}}$, $[\text{ADP}]_{\text{out}}$, $[P_{\text{in}}]_{\text{in}}$, n , pH_i , and pH_o . Some of these values have been chosen on the basis of reports on cytosolic estimations or computational predictions (31). We have generated a Windows-based software (E_{rev} estimator), in which the user can enter any value for any parameter participating in Eqs. 1–3, and observe the changes in $E_{\text{rev_ATPase}}$ and $E_{\text{rev_ANT}}$. The software and instructions on how to use it can be downloaded (<http://www.tinyurl.com/Erev-estimator>).

Directionality of the function of $\text{F}_0\text{F}_1\text{-ATPase}$ and ANT in isolated mitochondria during respiratory chain inhibition

To demonstrate that $\text{F}_0\text{F}_1\text{-ATPase}$ could indeed hydrolyze ATP without the assistance of the reverse operation of ANT in mitochondria, a prediction emerging from the model shown in Fig. 1, experiments were performed on isolated liver mitochondria compromised by targeted ETC inhibition. We measured $\Delta\Psi\text{m}$ and addressed the effect of cATR or oligomycin, specific ANT, and $\text{F}_0\text{F}_1\text{-ATPase}$ inhibitors, respectively, (Fig. 2). State 3 respiration in glutamate plus malate-supported mitochondria was initiated by ADP (1 mM) depolarizing mitochondria by $\sim 25 \text{ mV}$, and then complex I, III, or IV was inhibited by the specific inhibitors, rotenone (1 μM), stigmatellin (1.25 μM), and KCN (1 mM), respectively, which led to a clamp of $\Delta\Psi\text{m}$ to $\sim -100 \text{ mV}$. Subsequent addition of cATR (Fig. 2, black traces) caused a moderate repolarization in mitochondria inhibited by rotenone at complex I (Fig. 2A, D) and a minor depolarization when complex III or complex IV



E) Western blot of rat brain synaptic (syn), liver, brain nonsynaptic, and heart mitochondria for SUCLA2 and VDAC-1.

was inhibited by stigmatellin (Fig. 2B, D) or KCN, respectively (Fig. 2C, D). This result implied that at ~ -100 mV, the ANT was only minimally functional or it did not reverse, which is in accordance with the ADP-ATP steady-state exchange activity/ $\Delta\Psi_m$ relationship shown recently (13, 24). In contrast, when oligomycin (Fig. 2, red traces) was added instead of cATR, an immediate depolarization was observed, implying that ATPase was reversed. Oligomycin also caused an abrupt depolarization when added after cATR (Fig. 2D), indicating that F_0F_1 -ATPase could function in reverse, even when ANT is blocked. Lack of extramitochondrial ATP consumption under identical conditions is also shown in Supplemental Fig. 2D. Obviously, under the conditions shown in Fig. 2, ATP was available in the matrix from sources other than ANT. To address whether ATP-forming SUCL is present in mitochondria, we performed Western blots of rat liver, brain (synaptic and nonsynaptic), and heart mitochondria and observed SUCLA2 immunoreactivity using VDAC isoform 1 immunoreactivity as a reference in these mitochondria (Fig. 2E). In the experiments depicted in Fig. 2A–D, respiration of liver mitochondria was maintained by glutamate plus malate, which supports operation of SUCL toward the formation of succinate plus ATP.

If ATP generated by SUCL is critical for driving the reverse operation of F_0F_1 -ATPase, then mitochondria supported by substrates that favor the formation of succinyl-CoA and ADP in the SUCL-mediated reaction are expected to behave differently when challenged with ETC inhibitors. To test this hypothesis, we performed the experiments shown in Fig. 3. When succinate was given after cATR to mitochondria supported by glutamate and malate in the presence of rotenone (Fig. 3A, trace a), mitochondria were repolarized because of activation of the ETC *via* complex II. When ETC was inhibited distal from complex II by KCN (Fig. 3A, trace b) or stigmatellin (Fig. 3A, trace c), succinate induced an immediate loss of $\Delta\Psi_m$. Under this condi-

tion, succinate was shifted toward succinyl-CoA and ADP (plus P_i) formation, decreasing matrix ATP levels, and $\Delta\Psi_m$ could no longer be maintained in the absence of ANT reversal. Essentially similar results were obtained in mitochondria compromised by stigmatellin or KCN when succinate plus malonate were present together with glutamate and malate from the start of the experiments (Fig. 3B, traces b, d). In these conditions, malonate prevents succinate from being shuttled toward fumarate formation by succinate dehydrogenase, leaving no other option for succinate but to follow the succinyl-CoA ligase reaction toward succinyl-CoA and ADP (plus P_i) formation. In mitochondria where SUCL was partially inhibited by propionate (32) in the presence of stigmatellin or KCN, addition of cATR induced further depolarization (Fig. 3B, traces e and f *vs.* a and c, where no propionate was present), indicating that the decrease in SUCL function involves ANT reversal. With alternative substrates, such as succinate plus rotenone (Fig. 3C), β -hydroxy (OH)-butyrate (Fig. 3D) or β -OH-butyrate plus succinate and malonate (not shown), which do not support ATP generation in the SUCL-mediated reaction, $\Delta\Psi_m$ collapsed by both oligomycin and cATR added after inhibiting ETC by stigmatellin or KCN.

E_{rev_ATPase} and E_{rev_ANT} have been estimated for the various substrate combinations used for Figs. 2 and 3. In line with the above experiments, ATP efflux rates mediated by ANT were much slower with substrates not favoring substrate-level phosphorylation than with substrate combinations that allow substrate-level phosphorylation (see Supplemental Material).

Addressing the role of IF-1 on the directionality of F_0F_1 ATPase and ANT in isolated rabbit heart and liver mitochondria

There is a naturally occurring protein, termed inhibitor protein IF-1, that inhibits the consumption of ATP by

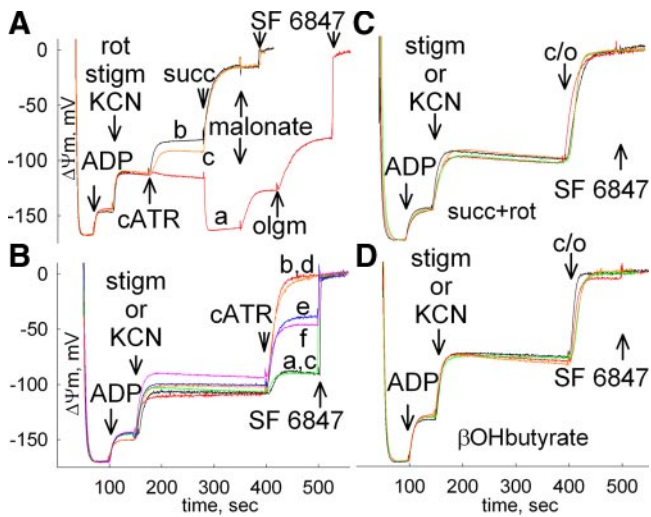


Figure 3. Effect of cATR or oligomycin (olgm) on $\Delta\Psi_m$ of mitochondria compromised at complex I, III, or IV in the presence of different substrate combinations. **A)** 1 mM ADP was given where indicated to mitochondria polarized by glutamate plus malate, and, subsequently, complex I, III, or IV was compromised by rotenone (rot, trace a), stigmatellin (stigm, trace c), or KCN (trace b), respectively; cATR was added where indicated. Subsequently, succinate (succ, 5 mM) was administered, followed by malonate (10 mM), where indicated. In trace a, oligomycin was also added after malonate. **B)** Mitochondria were allowed to be fully charged in the presence of the following substrate combinations: glutamate plus malate (traces a and c), glutamate plus malate plus succinate plus malonate (traces b and d), or glutamate plus malate plus propionate (traces e and f). Stigmatellin was given for traces a, e, and b; KCN was given in lieu of stigmatellin for traces c, f, and d. **C, D)** Mitochondria were polarized by succinate plus rotenone (**C**) or β -OH butyrate (**D**). ADP (1 mM) was given where indicated; subsequently, complex III or IV was compromised by stigmatellin or KCN, respectively; cATR or oligomycin was added where indicated (c/o). At the end of all experiments, 200 nM SF 6847 was added to achieve complete depolarization.

the F_0F_1 -ATPase (33). Such a mechanism would pose the obvious implication that on extensive mitochondrial depolarization, the ATPase would not reverse. This would lead to a lack of elevation of matrical [ADP] and a decrease in [ATP], disfavoring ANT reversal. To address the contribution of IF-1 in these conditions, we compared mitochondria from rabbit heart and rabbit liver with those from rat heart and rat liver. Rabbit heart exhibits the highest expression of IF-1 with the highest affinity for the ATPase, whereas rabbit liver exhibits very low expression of IF-1, almost similar to that of rat liver and heart, which exhibit low affinity for the ATPase (34–37). As shown in Supplemental Fig. 4, mitochondria from rabbit liver and heart were allowed to respire on glutamate plus malate, and ADP was given where indicated, while ATP efflux-influx rates (Supplemental Fig. 4A) and $\Delta\Psi_m$ (Supplemental Fig. 4C) were recorded. Subsequently, several 10 nM additions of SF 6847 were added, up to complete collapse of $\Delta\Psi_m$. From Supplemental Fig. 4A, it is apparent that rabbit liver mitochondria exhibited symmetric ATP production and consumption rates, as opposed to mitochon-

dria from rabbit heart, which exhibited diminished rates of ATP consumption compared with ATP production. That finding implied that ATP consumption processes were impaired, consistent with an inhibition of the ATPase by IF-1. This notion is further supported by the finding that the difference in $\Delta\Psi_m$ depolarization by SF 6847 in rabbit heart and liver mitochondria in the absence of oligomycin is greater than that in the presence of oligomycin (added after ADP), because in the presence of oligomycin, the effect of IF-1 is irrelevant (Supplemental Fig. 4C, D). Nevertheless, inhibition of the respiratory chain by rotenone, stigmatellin, or KCN (all given after ADP) did not cause a reversal of ANT in either mitochondria isolated from rabbit liver and heart, respiring on glutamate plus malate (Supplemental Fig. 5). From this finding, we inferred that the action of IF-1 would not confound the ANT reversal profile on inhibition of the respiratory chain.

Indeed, as shown in Supplemental Fig. 5, experiments comparing rabbit liver with rabbit heart mitochondria, performed similarly to those shown in Fig. 2A–C, exhibit the same pattern: addition of cATR to mitochondria inhibited at complex I, III, or IV, by rotenone, stigmatellin, or KCN, respectively, causes a small decrease in $\Delta\Psi_m$ or even repolarization, revealing slight or no reversal of the ANT. Addition of oligomycin in lieu of cATR caused an abrupt depolarization, consistent with a reversal of ATPase. However, addition of cATR on rabbit liver mitochondria inhibited by KCN showed an almost maximal depolarization (Supplemental Fig. 5F).

K_{ATP} channel activity in cell-attached mode as an ATP sensor assay during metabolic inhibition in cultured cortical neurons

Next, we wanted to address whether a reversal of F_0F_1 -ATPase without a parallel reversal of ANT suggested by our modeling (Fig. 1) and found in isolated mitochondria (Figs. 2 and 3) may operate also in *in situ* mitochondria in a cellular model. For this, we detected changes in the cytosolic ATP level by recording, in cell-attached mode, the activity of K_{ATP} channels, which open on a decrease in the cytosolic ATP level. We chose cortical neurons in culture, because they exhibit K_{ATP} channels (38–43). In this assay, plasmalemmal K_{ATP} channels were rendered hypersensitive to changes in the cytosolic ATP level by 0.1 mM pinacidil, a sensitizing compound (44). Creatine kinase was inhibited by the membrane-permeable inhibitor, iodoacetamide (0.25 mM), which is important for the K_{ATP} channel cytosolic ATP biosensor assay, because the enzyme buffers changes in cytosolic ATP and ADP levels. At 0.25 mM, creatine kinase of isolated brain mitochondria was completely inhibited (not shown). However, iodoacetamide is not specific for creatine kinase, it also inhibits several other enzymes (45). The unspecific targets are either too insensitive to inhibition at the concentration of iodoacetamide used in this study or among those that were appreciably inhibited that are unlikely to confound the outcome of our experiments

(45). As shown in **Fig. 4A**, application of either rotenone, stigmatellin, or NaCN alone failed to result in K_{ATP} channel activity (top traces). The holding potential of the pipette (V_p) was +20 mV, resulting in a plasma membrane potential of -60 to -80 mV in the cell-attached mode. This is because the plasma membrane potential of the cells before application of the mitochondrial toxins was in the -40- to -60-mV range (estimated by a current-clamp in the whole-cell mode). The cocktail used in the measurement contained inhibitors of inward-rectifying channels other than the K_{ATP} channel (detailed in Materials and Methods), which are not sensitive to [ATP] alterations (46). The presence of channel inhibitors also served the purpose of minimizing the Na^+ load and the cytosolic ATP consumption by the Na^+/K^+ ATPase. Subsequent application of the uncoupler decreased the plasma membrane potential even further, clamping it in the -10- to -30-mV range. Therefore, after addition of the uncoupler, the holding potential of the cell would be in the -30- to -50-mV range. Subsequent excision of the patch would result in holding the ruptured inside-out patch to +20 mV (therefore -20 mV outside), but that yielded very low single-channel current amplitudes, buried in the background noise. Elevation of the holding potential to +60 mV reproduced K_{ATP} channel activity, which is abundant owing to the absence of ATP from the extracellular medium; accordingly, while in the inside-out configuration, application of ATP closed the channels immediately. A gap-free recording at various holding potentials in the cell-attached configuration is shown in **Fig. 4B**. In **Fig. 4C**, a current-voltage relationship for K_{ATP} single-channel conductance is shown, exhibiting inward rectification. The inward rectification,

single-channel amplitude, burst-like openings, and sensitivity to ATP and glibenclamide provide enough assurance that the patches contained authentic K_{ATP} channels, and therefore the reporting channel activity was indeed a biosensor assay of cytosolic ATP levels. Collectively, the patch-clamp experiments in the cell-attached mode showed channel activity only in the presence of an uncoupler. ETC inhibition alone in the *in situ* mitochondria failed to induce plasmalemmal K_{ATP} channel activity, implying that the cytosolic ATP level, if altered, was below the detection level of the K_{ATP} channels. Because buffering of cytosolic ATP-ADP interconversions by creatine kinase was inhibited, there are two alternative but non-mutually excluding explanations for the lack of decline in the ATP level: 1) cytosolic ATP was not consumed by mitochondria with impaired ETC; and 2) mitochondria consumed cytosolic ATP, but a rebound increase in the glycolytic ATP output compensated adequately; therefore, the cytosolic ATP level remained unchanged. To check for a possible rebound increase in glycolytic ATP production, *i.e.*, a Pasteur effect, we investigated the acidification rates of the extracellular space in parallel with O_2 consumption rates in cortical cultures, which together reliably reflect the glycolytic activity.

Extracellular acidification rates, O_2 consumption, and $\Delta\Psi_m$ in cultured cortical neurons and astrocytes

Cultured cortical neurons and astrocytes were exposed to rotenone (2 μ M), stigmatellin (1.2 μ M), or NaCN (5 mM), with or without oligomycin (2 μ g/ml) for 3 min, and then the change in the ECAR (shown as normalized ΔpH) was measured in comparison with the basal

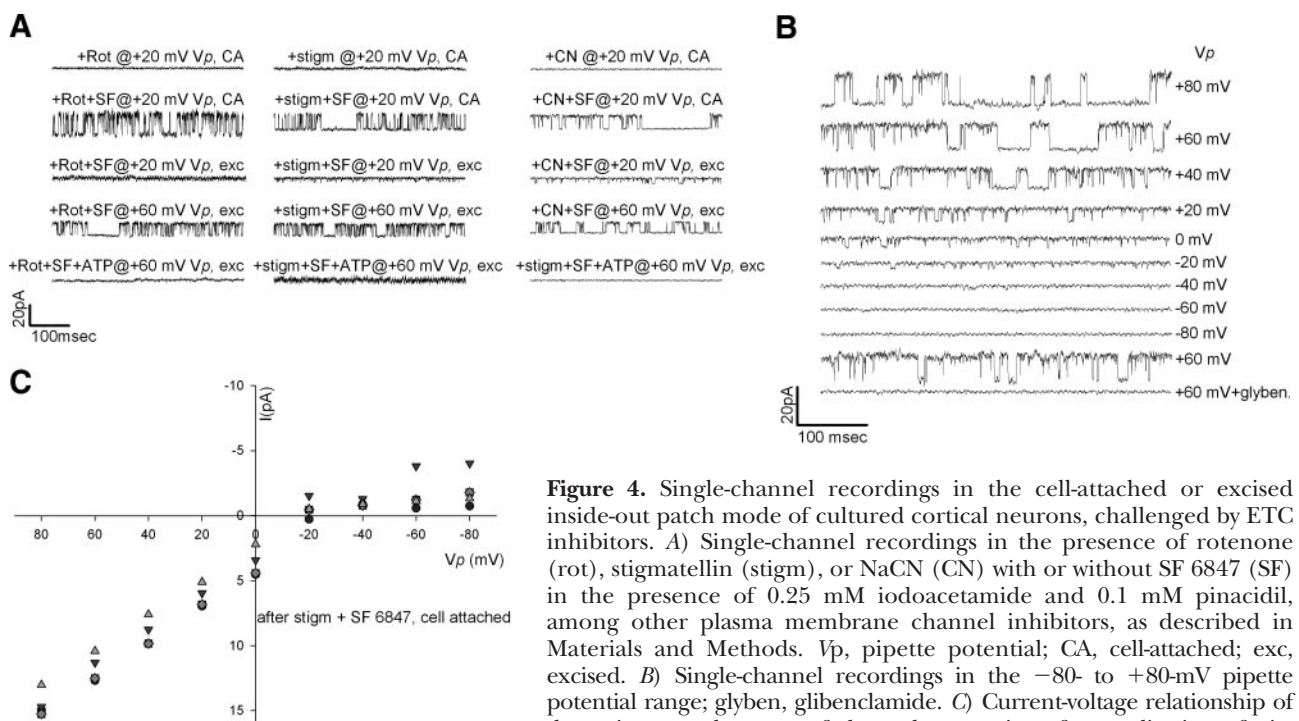


Figure 4. Single-channel recordings in the cell-attached or excised inside-out patch mode of cultured cortical neurons, challenged by ETC inhibitors. **A)** Single-channel recordings in the presence of rotenone (rot), stigmatellin (stigm), or NaCN (CN) with or without SF 6847 (SF) in the presence of 0.25 mM iodoacetamide and 0.1 mM pinacidil, among other plasma membrane channel inhibitors, as described in Materials and Methods. V_p , pipette potential; CA, cell-attached; exc, excised. **B)** Single-channel recordings in the -80- to +80-mV pipette potential range; glyben, glibenclamide. **C)** Current-voltage relationship of the unitary conductance of channels appearing after application of stigmatellin and SF 6847 in the cell-attached mode.

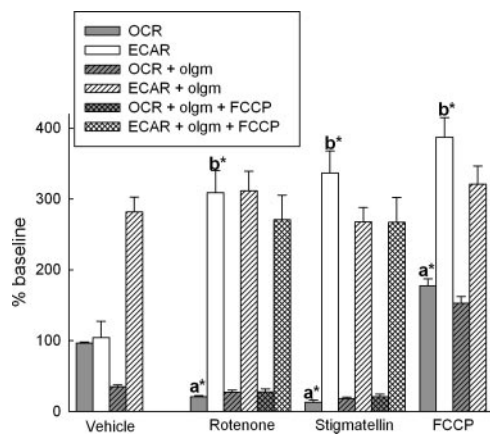


Figure 5. Effect of rotenone or stigmatellin on oxygen consumption and extracellular acidification rates in rat cortical cultures with or without oligomycin (olgm). OCR and ECAR were determined in a microplate format respirometry/pH assay. Solid bars indicate mean $\pm$ SE of OCR (gray) and ECAR (white), normalized to resting conditions, after mixing of vehicle, rotenone (2 μ M), stigmatellin (1.2 μ M), or FCCP (1 μ M) to the assay medium. Hatched bars indicate the above conditions in the presence of oligomycin (2 μ g/ml) and cross-hatched bars after addition of FCCP (1 μ M). Data are from 4 independent cell culture preparations; $n = 10$ –12 wells/condition. * $P < 0.05$ vs. vehicle for OCR (a) or ECAR (b); ANOVA with Tukey's *post hoc* test.

rates (Fig. 5). All of these tested inhibitor combinations, except for cyanide, triggered a 150–250% increase in the acidification rates. In the presence of cyanide, a gradual, but significant, alkalization of the low-buffering-capacity medium (optimized for the ECAR assay) was found, which may be the result of the volatility of HCN (not shown).

ECAR results from glycolytic lactate and mitochondrial carbonic acid production (47), which is typical in the case of carbonyl cyanide 4-(trifluoromethoxy)phenylhydrazone (FCCP). However, in the presence of oligomycin or rotenone/stigmatellin, when the respiration is strongly suppressed (Fig. 5), acidification is mainly due to glycolysis. Rotenone or stigmatellin did

not trigger stronger acidification than oligomycin, suggesting that either mitochondria do not use cytosolic ATP or glycolysis runs at maximal capacity under these conditions. FCCP added after rotenone or stigmatellin did not trigger further acidification, supporting the latter scenario but not negating the former. Oligomycin strongly attenuated basal respiration, suggesting that the basal proton leak is very small: (oligomycin – stigmatellin)/(basal – stigmatellin) = (35 – 20.5)/(100 – 20.5) = 18.2%; thus, the basal ATP turnover is accountable for 81.8% of the mitochondrial respiration. This small proton leak is expected to be further decreased when $\Delta\Psi_m$ is partially depolarized in the presence of rotenone (48).

In separate experiments, $\Delta\Psi_m$ was determined in intact cultured cortical neurons and astrocytes using fluorescence microscopy. TMRM and PMPI fluorescence of whole cell bodies was followed in time, and $\Delta\Psi_m$ was determined by deconvoluting the fluorescence intensity time lapses, taking into account the slow, potential-dependent redistribution of the probes. Rotenone (2 μ M) or stigmatellin (1.2 μ M) depolarized *in situ* mitochondria in cortical neurons to ~ -85 mV from the assumed resting potential (-150 mV) (49), and subsequent addition of oligomycin (2 μ g/ml) induced further depolarization (Fig. 6A, B). In contrast, the ANT inhibitor bongkreikic acid (BKA; 50 μ M; cATR was not used because of its inability to penetrate the cell membrane) first transiently increased mitochondrial polarization, which was followed by a delayed depolarization similar to the level observed with oligomycin. Importantly, depolarization in the presence of BKA exhibited asynchrony between cells and 24% of BKA-treated cells retained the more polarized $\Delta\Psi_m$ for the next 30 min ($n = 233$). In contrast, oligomycin-treated cells depolarized synchronously.

The BKA-evoked transient repolarization of the $\Delta\Psi_m$ in the presence of rotenone or stigmatellin can be explained by the alleviation of ATP turnover on a forward working ANT or by a decrease in proton leak. Mitochondria of BKA- and oligomycin-treated neurons

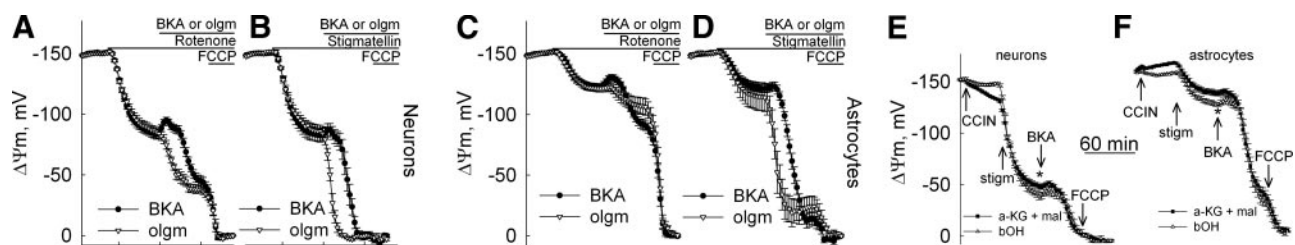


Figure 6. Effect of BKA, oligomycin (olgm), and a metabolic condition not supporting succinate thiokinase reaction on the rotenone- or stigmatellin-evoked depolarization of $\Delta\Psi_m$ in cultured rat cortical neurons (A, B, E) and astrocytes (C, D, F). $\Delta\Psi_m$ was followed in intact cells with fluorescence microscopy using the potentiometric probes PMPI and TMRM. $\Delta\Psi_m$ (mV) was calculated by a compartment model-based, dynamic calibration of single-cell fluorescence intensities (see Supplemental Material). A–D) BKA (50 μ M; ●), oligomycin (2 μ g/ml; ▽), rotenone (2 μ M), stigmatellin (1.2 μ M), and FCCP (1 μ M) were added by mixing into the assay medium containing 15 mM glucose and were present as indicated by bars. E, F) CCIN (500 μ M) was added together with (R)- β -OH-butyrate (5 mM; ■) or with α -ketoglutarate plus L-malate (both 5 mM; ▴) in the presence of 5 mM glucose, as marked (CCIN). Following additions were performed as described above. * $P < 0.05$; Student's *t* test. Data were pooled from 3 to 5 independent cell culture preparations; data points represent the means $\pm$ SE of $n = 6$ –9 view fields containing 200–300 neurons or 80–150 astrocytes per condition.

had essentially the same $\Delta\Psi_m$ [-147.9 ± 2.4 mV, $n=6$ view fields (196 cells) and -146.3 ± 3.2 mV, $n=5$ (179 cells), respectively; see Supplemental Figs. 6 and 7]. In separate experiments, the effect of oligomycin (2 μ g/ml) or BKA (50 μ M) on the respiration of cortical neurons was also identical; the basal respiration was decreased to $35.8 \pm 3.7\%$ ($n=8$ wells) and $35.9 \pm 3.9\%$ ($n=6$ wells) of the control, respectively, as measured ~ 10 min after drug addition using the XF24 Extracellular Flux Analyzer. Both in the presence of oligomycin and of BKA, mitochondria were in state 4 with identical $\Delta\Psi_m$ and respiration rates; therefore, the proton leak was identical and could not be responsible for the repolarization observed with BKA. Moreover, in the presence of ETC inhibition, when $\Delta\Psi_m$ is diminished, the extent of contribution of the proton leak to $\Delta\Psi_m$ is 3–12 times smaller than that obtained in fully charged mitochondria (48). The existence of a subpopulation of cells that maintained the repolarized $\Delta\Psi_m$ in the presence of rotenone or stigmatellin plus BKA for a longer period of time suggests that these cells are able to maintain $\Delta\Psi_m$ without hydrolyzing cytosolic ATP. Assaying astrocytes that were present in the cortical cultures for $\Delta\Psi_m$ yielded qualitatively similar results (Fig. 6C, D). Next, mitochondrial pyruvate uptake was inhibited by α -cyano-4-hydroxy-cinnamate (500 μ M), and (*R*)- β -OH-butyrate (5 mM) or α -ketoglutarate (5 mM) plus L-malate (5 mM) was provided as a mitochondrial substrate in the presence of 5 mM glucose. The first combination does not support ATP generation in the SUCL-mediated reaction (as shown above in Fig. 3C), and

this resulted in a significantly larger drop of $\Delta\Psi_m$ on addition of stigmatellin (Fig. 6E, F, *) than with the latter combination, by 8.6 ± 6.6 mV in neurons and by 10.7 ± 4.4 mV in astrocytes (see raw fluorescence data and additional control experiments in Supplemental Figs. 6 and 7).

Cytosolic and nuclear ATP pools during ETC inhibition in COS-7 cells

To further test the hypothesis that extramitochondrial ATP depletion may not occur during ETC inhibition, we transfected COS-7 cells with plasmids encoding firefly luciferases that target the cytosol or nucleus and performed a bioluminescence assay of intracellular [ATP] originating from the respective compartments during metabolic inhibition. COS-7 cells were used because of their high transfection efficiencies compared with those obtained from primary cells probed in different experiments, as detailed above. The targeting of the luciferases was specific for the intended compartments (Fig. 7A, B, insets). All experiments were performed in the presence of 0.25 mM iodoacetamide to block buffering and spatial transmission of phosphorylation potential (50). Two metabolic conditions were considered: supplying cells with glucose so that both glycolysis and mitochondria contributed to cytosolic and nuclear ATP pools; and replacing glucose with 2-deoxyglucose plus pyruvate, a condition in which only mitochondria maintained ATP production. As shown in Fig. 7A, in the presence of glucose, the addition of

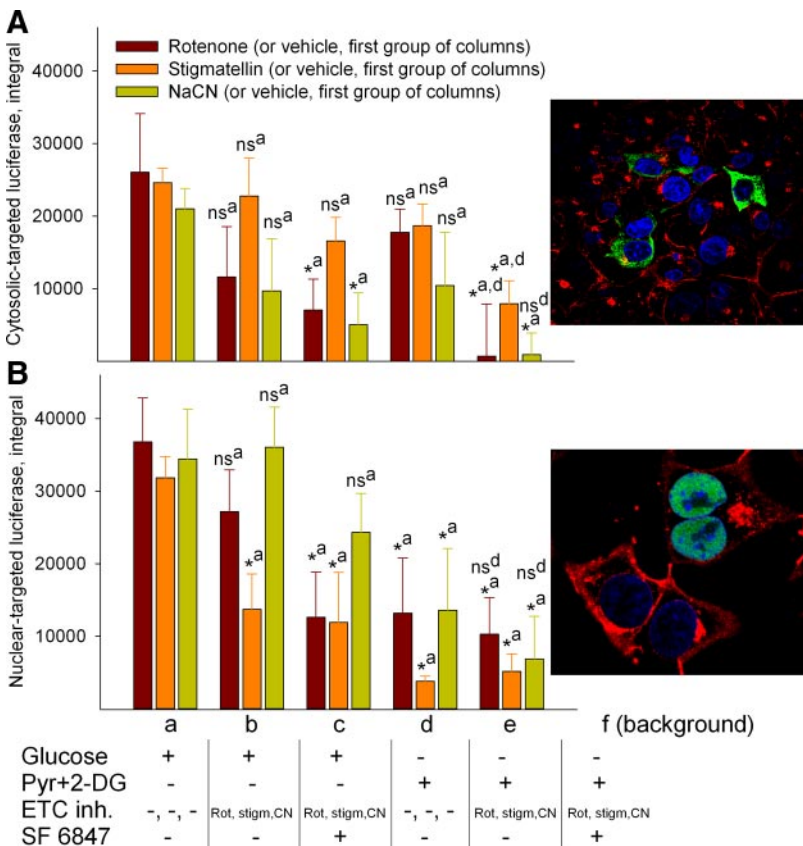


Figure 7. Integrals of luciferin/luciferase-assisted ATP luminescence in COS-7 cells during various metabolic conditions, transfected with cytosolic- or nuclear-targeted luciferases. A) Integral of cytosol-targeted firefly luciferase luminescence signal for various metabolic conditions, as shown at bottom (a–e). Glucose (15 mM); Pyr, pyruvate (5 mM); 2-DG, 2-deoxyglucose (2 mM); ETC inh, ETC inhibitor. B) Same as in A, but for nuclear-targeted luciferase. For background subtraction, the following metabolic condition (f) was used in both panels: no glucose + SF 6847 + ETC inhibitor + pyruvate + 2-deoxyglucose. Data were pooled from 3 independent cell culture preparations. * $P < 0.05$ vs. vehicle; ANOVA with Tukey's *post hoc* test compared with the vehicle treatment. a, glucose only; d, Pyr + 2DG only; ns, nonsignificant. Insets: confocal micrographs (0.65- μ m z sections) of luciferase targeted to the cytosol (top) or to the nucleus (bottom), stained by Alexa 647 agglutinin (before permeabilization) that decorates the plasma membrane and the Golgi apparatus (red), DAPI for the nucleus (blue), and antibodies raised against the luciferase (green).

rotenone or NaCN decreased cytosolic ATP level, but this failed to reach statistical significance. The concomitant presence of the uncoupler SF 6847 led to a further drop in $[ATP]_{\text{cyt}}$ that reached statistical significance. However, in the absence of glucose, when glycolysis is unable to mount a Pasteur effect, the cytosolic ATP level was dropped by the inhibitors. The concomitant presence of the uncoupler SF 6847 led to a further drop of $[ATP]_{\text{cyt}}$ (not shown), and this condition served as the background for baseline subtraction of the luciferase signal. Considering $[ATP]_{\text{nuc}}$ (Fig. 7B), the addition of either rotenone or NaCN did not result in a significant drop in $[ATP]_{\text{nuc}}$ levels compared with the absence of the ETC inhibitor, but the effect of stigmatellin was significant. The decrease in $[ATP]_{\text{nuc}}$ under conditions of glycolysis plus uncoupling or the presence of an ETC inhibitor or absence of glycolysis is in accord with the interpretation of results obtained by Gajewski *et al.* (27), showing that the $[ATP]_{\text{nuc}}$ pool may become depleted faster than the $[ATP]_{\text{cyt}}$ pool. This finding could suggest that in intact cells certain compartments are in preferential contact with mitochondria for the transfer of phosphorylation potential. From the above experiments we conclude that in COS-7 cells inhibition of the respiratory chain may not lead to uniform consumption of cytosolic and/or nuclear ATP.

DISCUSSION

In the present study, we demonstrated that during respiratory chain inhibition or submaximal uncoupling, when mitochondria not only stop synthesizing ATP but also work as ATP consumers, F_0F_1 -ATPase is able to work in reverse without the concomitant reversal of ANT. This phenomenon was demonstrated on 3 levels. First, computational modeling (Fig. 1) suggested that changes in $E_{\text{rev_ANT}}$ and in $E_{\text{rev_ATPase}}$ define a $\Delta\Psi_m$ range (B space), in which F_0F_1 -ATPase runs in reverse, whereas the forward function of ANT is maintained. Under these conditions, $\Delta\Psi_m$ supported by the ATP-hydrolyzing F_0F_1 -ATPase is in a range more negative than the $E_{\text{rev_ANT}}$, and thereby ANT would not reverse. Second, experiments with isolated mitochondria (Figs. 2 and 3) indicated that this phenomenon could indeed exist when respiration of mitochondria is impaired or submaximally uncoupled (Supplemental Material) and suggested that matrix substrate-level phosphorylation could be critical in providing ATP for the reversed F_0F_1 -ATPase. Third, data obtained with cultured cells (Figs. 4–7) support the suggestion that *in situ* mitochondria are also likely to exhibit reversal of F_0F_1 -ATPase but not of ANT, although the contribution of ATP provision by *SUCLA2*-mediated substrate level phosphorylation *in situ* has not been pinpointed. It is clear from both the model and the experiments with *in vitro* and *in situ* mitochondria that when depolarization is severe enough, *i.e.*, when $\Delta\Psi_m$ is less negative than the $E_{\text{rev_ANT}}$, a condition observed in the presence of adequate amount of an uncoupler, mitochondria could

consume extramitochondrial ATP translocated from the cytosol by the ANT.

Cytosolic ATP consumption by mitochondria under impaired respiration was reported for isolated nerve terminals challenged with rotenone and FCCP (51). This phenomenon is also characteristic for oxidative stress (52) when inhibition of α -ketoglutarate dehydrogenase (53), a key Krebs cycle enzyme, limits the NADH availability and, as a result, the respiratory function of mitochondria. A novel concept emerging from our study is that matrix substrate-level phosphorylation could be an endogenous rescue mechanism supporting the reverse operation of F_0F_1 -ATPase, with the benefit of maintaining $\Delta\Psi_m$ at a suboptimal level, where mitochondria are depolarized, but not sufficiently for the ANT to reverse. This would assist in preserving the ATP pool produced by glycolysis, which is evidently crucial for the survival chances of cells due to maintenance of the function of the vital ATP-dependent transporters, such as Na^+/K^+ -ATPase and Ca^{2+} -ATPase. Cells could be saved as long as matrix substrate-level phosphorylation is able to provide ATP for the F_0F_1 -ATPase in impaired mitochondria. The importance of succinate thiokinase can be better appreciated in the severe phenotype observed in patients suffering from *SUCLA2* (or *SUCLG1*) deficiencies (54). However, it would be difficult to attribute the phenotype exclusively to the provision of ATP by succinate thiokinase *vs.* other very important metabolic branches, such as the maintenance of the citric acid cycle (20), ketone metabolism (16), or heme synthesis (21). Nonetheless, it is noteworthy that increasing mitochondrial substrate-level phosphorylation can rescue respiratory growth of an ATP synthase-deficient yeast (55). It is also noteworthy to mention that recent experiments by Balaban's group (22) pointed to succinyl-CoA synthetase as a target activated by matrix P_i , contributing to the anaerobic provision of ATP in mitochondria. This finding would obviously have the implication of providing ATP for the consuming F_0F_1 -ATPase, thereby maintaining a moderate $\Delta\Psi_m$ that, if more negative than $E_{\text{rev_ANT}}$, could contribute to sparing of cytosolic ATP pools. Likewise, a beneficial by-product is the provision of matrix P_i from matrix ATP hydrolysis, which could activate succinyl-CoA synthetase toward extra ATP formation. The finding of Balaban's group strongly supports our concept, suggesting the importance of succinyl-CoA synthase in providing ATP in impaired mitochondria.

Furthermore, our results are in line with and can explain the repeatedly reported observation that inhibition of the respiratory chain of *in situ* mitochondria failed to activate in all cells the cytosolic ATP sensing mechanisms activated by a drop in $[ATP]$ levels (44, 56–58). It may be possible that the energy depletion protocols used did not permit ANT reversal in all cells. Accordingly, application of uncouplers to cells invariably led to opening of plasmalemmal K_{ATP} channels, in line with sufficient loss of $\Delta\Psi_m$, leading to ANT reversal (44).

The ability of mitochondria to enter the B space (Fig. 1) could be essential for the survival of cells when

respiration is impaired in some but not all *in situ* mitochondria. This scenario is known to exist in pathological conditions, when the mitochondrial DNA (mtDNA) pool of individual cells is heteroplasmic (59–62). It is reasonable to assume that the substrate-level phosphorylation by SUCL in respiration-deficient mitochondria could act as a quality-control switch mechanistically involved in gene shifting of heteroplasmic mtDNA toward the wild type.

Finally, the significance of our findings provides a further reason for focusing research efforts on mitochondrial damage, which is observed in a variety of currently untreatable diseases: it has been well established, that a breach in inner mitochondrial integrity occurs during the activation of the so-called permeability transition pore (PTP) (63, 64). Opening of this pore allows mitochondria to hydrolyze cytosolic ATP (14), and along with the release of proapoptotic proteins this has been implicated in the etiology of a number of diseases (65, 66). Therefore, the beneficial effects of targeted inhibition of PTP offer the possibility of mitochondria with the capacity of matrix substrate-level phosphorylation to maintain $\Delta\Psi_m$ in the range where ANT reversal is prevented, thereby ameliorating cytosolic ATP consumption and avoiding cell death. **[F]**

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